Response to Reviewers — Second Cut

Team 3 · *Servo Characterization for the SOFIA Modular Kinetic Surface*

Acknowledgements

The authors thank the reviewers for their careful evaluation of the first-cut manuscript. <One sentence on how the comments improved the paper.> All comments were considered; the responses below follow the keys of the review report (R1.1, R1.2, . . .). Line, section, figure, table, and equation numbers refer to the revised manuscript that accompanies this letter.

Main changes. (i) <Main change 1> (Section I). (ii) <Main change 2>. (iii) <Main change 3>.

Summary of changes (observation–response–change matrix)

17 observations: 15 addressed, 2 partially addressed, 0 not addressed (justified).

Key	Status	Location in the revised paper
R1.1	**[Addressed]**	Title and author block; Section I
R1.2	**[Addressed]**	Table I, Table II, and Table III
R1.3	**[Addressed]**	Table II
R1.4	**[Addressed]**	Table I, Table II, and Fig. 1
R1.5	**[Partially addressed]**	Sections IV, V, and VI
R1.6	**[Addressed]**	Manuscript preamble and document structure
R2.1	**[Addressed]**	Abstract and IEEE Keywords
R2.2	**[Partially addressed]**	Section II and Table III
R2.4	**[Addressed]**	lines 36-47
R3.1	**[Addressed]**	Fig. 1
R3.3	**[Addressed]**	lines 193-199, Table VI, (1)
R3.5	**[Addressed]**	Table IV, Table V,Table VI
R3.6	**[Addressed]**	Table VIII
R3.7	**[Addressed]**	Table IX,Table X,Table XI, Table XII
R3.8	**[Addressed]**	Table 1
R4.1	**[Addressed]**	lines 193-199, Table VI, (1)
R4.2	**[Addressed]**	Section I

Reviewer 1 — *Format, LaTeX and template compliance*

R1.1 [Addressed]

Reviewer comment: El título es “LAB-DST SOFIA O26”, que no describe el trabajo. No hay afiliación en [1] El apellido del profesor aparece como “Gonzales Jimenez”, y el encabezado dice “Technical Datasheet of SOFIA Demonstrator Module”.

Response: We revised the manuscript title, author information, institutional affiliation, and running header to follow the IEEE journal template and to reflect the scope of servo characterization. The revised title is “Servo Characterization for the SOFIA Modular Kinetic Surface.”

Changes in the revised paper: Title and author block; Section I

R1.2 [Addressed]

Reviewer comment: Las tablas de requisitos, de la ESP32-C3, del PCA9685, del MG90S y de verificación no tienen no se citan en el texto; aparecen una tras otra sin párrafo que las introduzca. Las fuentes se colocan como líneas sueltas (“Source: ESP32-C3 Datasheet, Espressif Systems.”, Línea 105).

Response: We reviewed the table captions, labels, and in-text cross-references. The revised manuscript refers to the comparative matrix, benchmarking table, and system requirements using LaTeX labels and references rather than manually entered table numbers.

Changes in the revised paper: Table I, Table II, and Table III

R1.3 [Addressed]

Reviewer comment: En la tabla de benchmarking el texto de las columnas se encima (p. ej. "Continuous (non-discrete)" invade la columna Modularity).

Response: We reformatted the benchmarking table to fit the IEEE two-column layout. The revised table uses a column-width-constrained tabularx environment, adjusted column spacing, and smaller text to improve readability and prevent horizontal overflow.

Changes in the revised paper: Table II

R1.4 [Addressed]

Reviewer comment: Todos los flotantes usan [H] (paquete float).

Response: We revised the figure and table environments and their placement options to better follow IEEE formatting conventions. Wide comparative content uses a two-column table environment, while the remaining figures and tables use single-column floats as appropriate. Final placement remains subject to LaTeX's float algorithm.

Changes in the revised paper: Table I, Table II, and Fig. 1

R1.5 [Partially addressed]

Reviewer comment: Faltan las secciones Results, Discussion y Conclusions (al menos como encabezados) y todas las secciones de cierre.

Response: We added the Results, Discussion, and Conclusions sections to establish the required manuscript structure. The Results section also includes subsections for the experimental setup, servomotor dynamic response, and time-constant estimation. These sections are not yet populated with experimental findings, so the observation is only partially addressed. No experimental results are claimed at this stage.

Changes in the revised paper: Sections IV, V, and VI

R1.6 [Addressed]

Reviewer comment: Al final del archivo hay una tabla comentada con valores de ejemplo ("Manufacturer Y, City, Country", "Continuousrotation servo, rated torque T at V V"), copiada del ejemplo de la clase de BOM. En figures/ hay cinco imágenes (nop (1..5).png) que no se usan, con espacios en el nombre.

Response: We cleaned up the manuscript structure and removed unused template material while retaining the IEEEtran journal document class and the packages required by the paper's tables, figures, equations, and algorithm.

Changes in the revised paper: Manuscript preamble and document structure

Reviewer 2 — *Writing and structure*

R2.1 [Addressed]

Reviewer comment: El abstract y las palabras clave eran el texto guía de la plantilla; en esta versión quedaron vacíos.

Response: We rewrote the abstract to state the system context, the research problem, the proposed experimental methodology, and the scope of the work. The abstract now specifies that the complete SOFIA installation comprises 16 modules and 32 actuators, while the proposed experimental characterization is restricted to the two servomotors of one removed module. We also updated the IEEE Keywords to match the paper's technical focus.

Changes in the revised paper: Abstract and IEEE Keywords

R2.2 [Partially addressed]

Reviewer comment: Materials and Methods se presenta como otro documento: "The objective of this document is to verify the compatibility of the selected components" y "The document structure follows the LabDST documentation requirements for technical datasheets". Es la ficha técnica de la clase de hardware insertada tal cual.

Response: We reorganized the opening of Materials and Methods around three stages: hardware configuration, experimental measurement, and dynamic-response analysis. The revised text describes the use of a removable crown-shaped fixture with limit switches to detect angular reference positions and identifies relevant measurement conditions, including commanded angular displacement, direction of motion, supply voltage, and elapsed time. It also clarifies that only two servomotors from one SOFIA module will be characterized. However, the complete measurement sequence, number of repetitions, data-acquisition procedure, and model-identification criteria remain to be specified; therefore, this observation is partially addressed.

Changes in the revised paper: Section II and Table III

R2.4 [Addressed]

Reviewer comment: Salvo un párrafo final, Materials and Methods está formada solo por tablas; las subsecciones A–F no tienen texto que explique qué muestra cada tabla ni qué se concluye.

Response: We rewrote the introduction and all mentions of servo characterization to explicitly state that it is the proposed contribution of this work. We simplified the work to only have the servo characterization as our only differentiation. (lines 36-47). The revised text now reads:

> *"This work proposes an experimental method to estimate individual servo time constraints with a mechanical solution"*

Changes in the revised paper: lines 36-47

Reviewer 3 ***Technical content***

R3.1 [Addressed]

Reviewer comment: No hay arquitectura preliminar del sistema en figura: cómo se conectan la cámara y el PC, la ESP32-C3, los dos PCA9685, los 32 servos, los transceptores RS-485 y la alimentación

Response: Added a System Architecture Diagram Fig. 1 showing the host, ESP32-C3, dual PCA9685 drivers, 32 servos, RS-485 transceivers. For the servo mechanical time constant an external module will be designed for measuring every servo motor.

Changes in the revised paper: Fig. 1

R3.3 [Addressed]

Reviewer comment: La "constante de máquina" o "mechanical time constant" del servo no se define: no hay modelo, variables ni ecuaciones. El diferenciador promete compararla contra el "valor teórico" del MG90S, que la hoja de datos no reporta.

Response: Formulated the first-order dynamic model $\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m}$ (1) and clarified that the reference baseline is derived from the manufacturer's no-load speed specification ($0.10\ \mathrm{s}/60^\circ$ at 4.8 V) Table VI.

Changes in the revised paper: lines 193-199, Table VI, (1)

R3.5 [Addressed]

Reviewer comment: Las fichas técnicas tienen solo parámetro, valor y verificación. No indican la versión ni la página de la hoja de datos, ni justifican el parámetro frente al requerimiento. El par del MG90S aparece como "1.8–2.2 kgf cm" sin decir a qué voltaje corresponde cada valor.

Response: Added source and justifications to the datasheets and updated the MG90S table to specify torque values at both 4.8V ($1.8\ \mathrm{kgf \cdot cm}$) and 6.0V ($2.2\ \mathrm{kgf \cdot cm}$).

Changes in the revised paper: Table IV, Table V,Table VI

R3.6 [Addressed]

Reviewer comment: El BOM no distingue la cantidad por unidad de la del sistema. Lista 1 ESP32-C3 y 2 PCA9685 pero 16 transceptores RS-485, sin explicar la arquitectura que eso implica. Usa "3D Printed" como fabricante, contiene elementos poco claros ("Pro3" de Tower Pro, "Axis 1 9mm x 9mm", "Inser M3x5x4") y lista conectores JST-PH, cuando la arquitectura de SOFIA usa JST-VH. Tampoco tiene descripción técnica ni localidad.

Response: Updated the BOM table to distinguish unit vs. system quantities, add descriptions for the parts, set the manufacturer to "In-house (LabDST, ITESO)", and added manufacturing origin.

Changes in the revised paper: Table VIII

R3.7 [Addressed]

Reviewer comment: La matriz de alternativas tiene un solo criterio por función, sin ponderación ni calificación.

Response: Updated the alternative matrix to include 1 table for every component stating features and limitations of the alternatives for the SOFIA.

Changes in the revised paper: Table IX,Table X,Table XI, Table XII

R3.8 [Addressed]

Reviewer comment: El algoritmo usa r0 para la posición del módulo y ^v para la dirección, en lugar de la notación del curso (r0,2, ^r2,p). Además escribe atan2, asin y Clamp en itálica matemática, sin operatorname ni Call.

Response: Updated math operators to standard notation (atan2, arcsin, Clamp) and explicitly labeled the trajectory step as an Exponential Moving Average (EMA)

Changes in the revised paper: Table 1

Reviewer 4 — *D-Lab and state of the art*

R4.1 [Addressed]

Reviewer comment: El diferenciador propone asignación automática de direcciones Modbus y caracterización de la constante de máquina de cada servo. El artículo menciona ambas en la introducción, pero se enfoca en justificar la arquitectura electrónica (Línea 83) y no desarrolla ninguna de las dos.

Response: Formulated the first-order dynamic model $\theta(t) = \theta_f + (\theta_0 - \theta_f)e^{-t/\tau_m}$ (1) and clarified that the reference baseline is derived from the manufacturer's no-load speed specification ($0.10\ \mathrm{s}/60^\circ$ at 4.8 V) Table VI. We simplified the article to only use the servo mechanical time constant as differentiation

Changes in the revised paper: lines 193-199, Table VI, (1)

R4.2 [Addressed]

Reviewer comment: No hay planteamiento formal del problema, pregunta de investigación ni objetivos.

Response: We added a formal Problem Statement, Research Question, and Objectives in natural narrative format at the end of the Introduction (Section I), addressing per-servo time-constant estimation and actuation synchronization. The revised text now reads:

Changes in the revised paper: Section I

"Standard low-cost micro-servos exhibit manufacturing tolerances that result in non-uniform actuation speeds and transient response times across different units. In a 16-unit, 32-actuator modular kinetic surface like SOFIA, driving all servos with uniform control signals creates dynamic motion desynchronization and mechanical stress during normal use. To address this limitation, this work investigates the following research question: Can experimental estimation of individual servo mechanical time constants (τ_m) and their integration into a trajectory controller improve dynamic positional synchronization across a 32-actuator kinetic surface? This encompasses three primary objectives: (1) designing a PCB integrating an ESP32-C3 microcontroller, dual PCA9685 drivers, and RS-485 interfaces; (2) developing an automated micro-test protocol to capture step-response and extract individual time constants; and (3) integrating individual compensation parameters into the surface tracking controller to equalize dynamic settling times across all actuators."